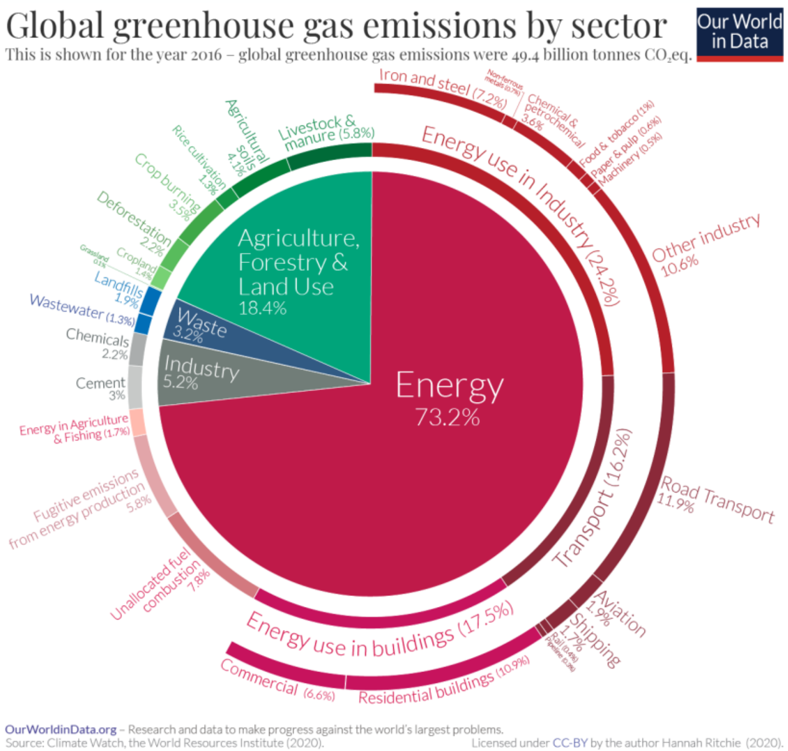
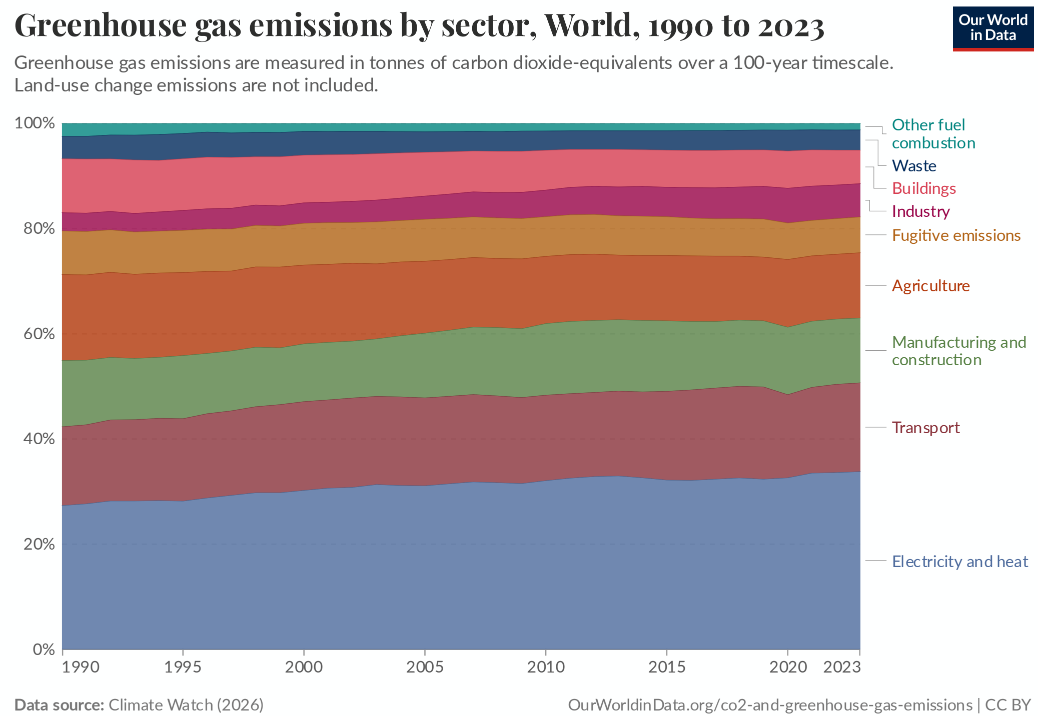
Sustainable Energy Technologies

Sustainable transport

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

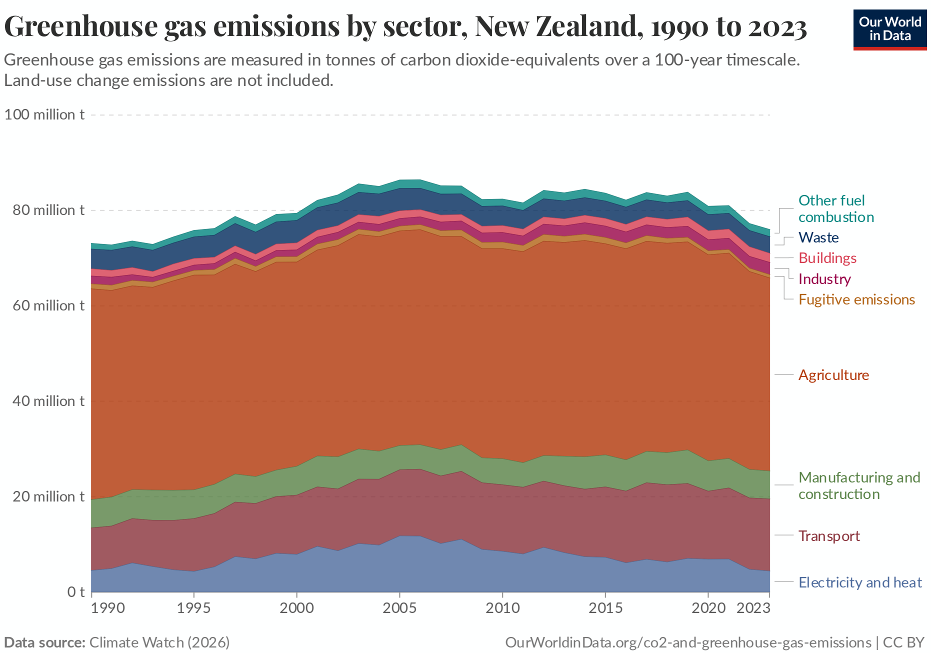
Transport sector responsible for 16%+ of GHG

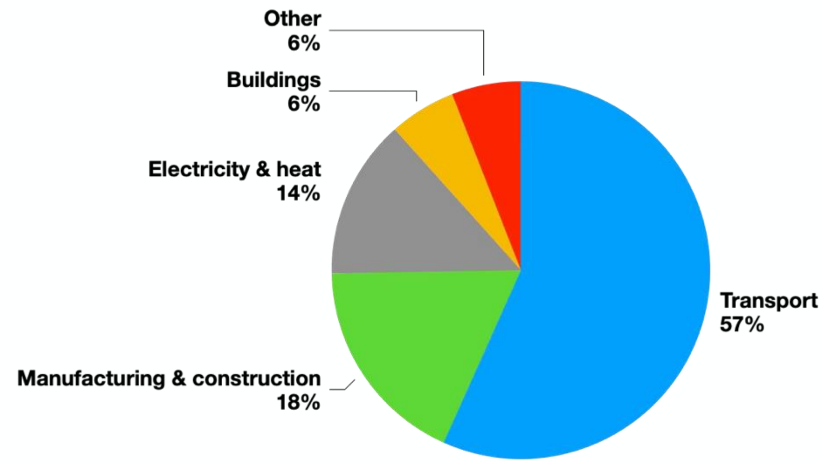




Sector by sector: where do global greenhouse gas emissions come from? https://ourworldindata.org/emissions-by-sector 2016

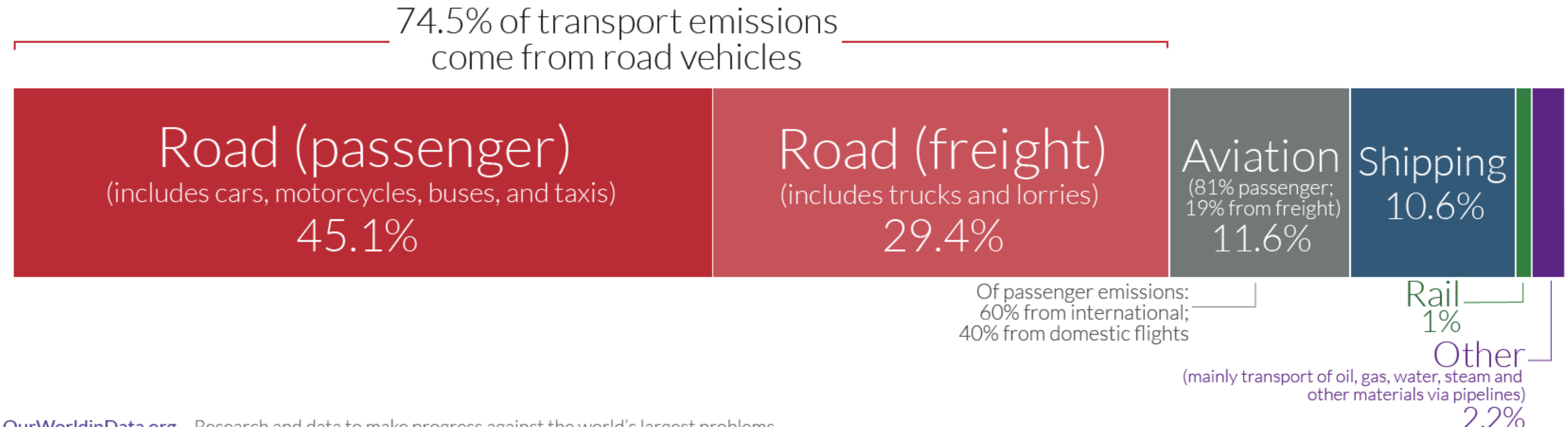
Transport sector responsible for ~20%+ of GHG in NZ





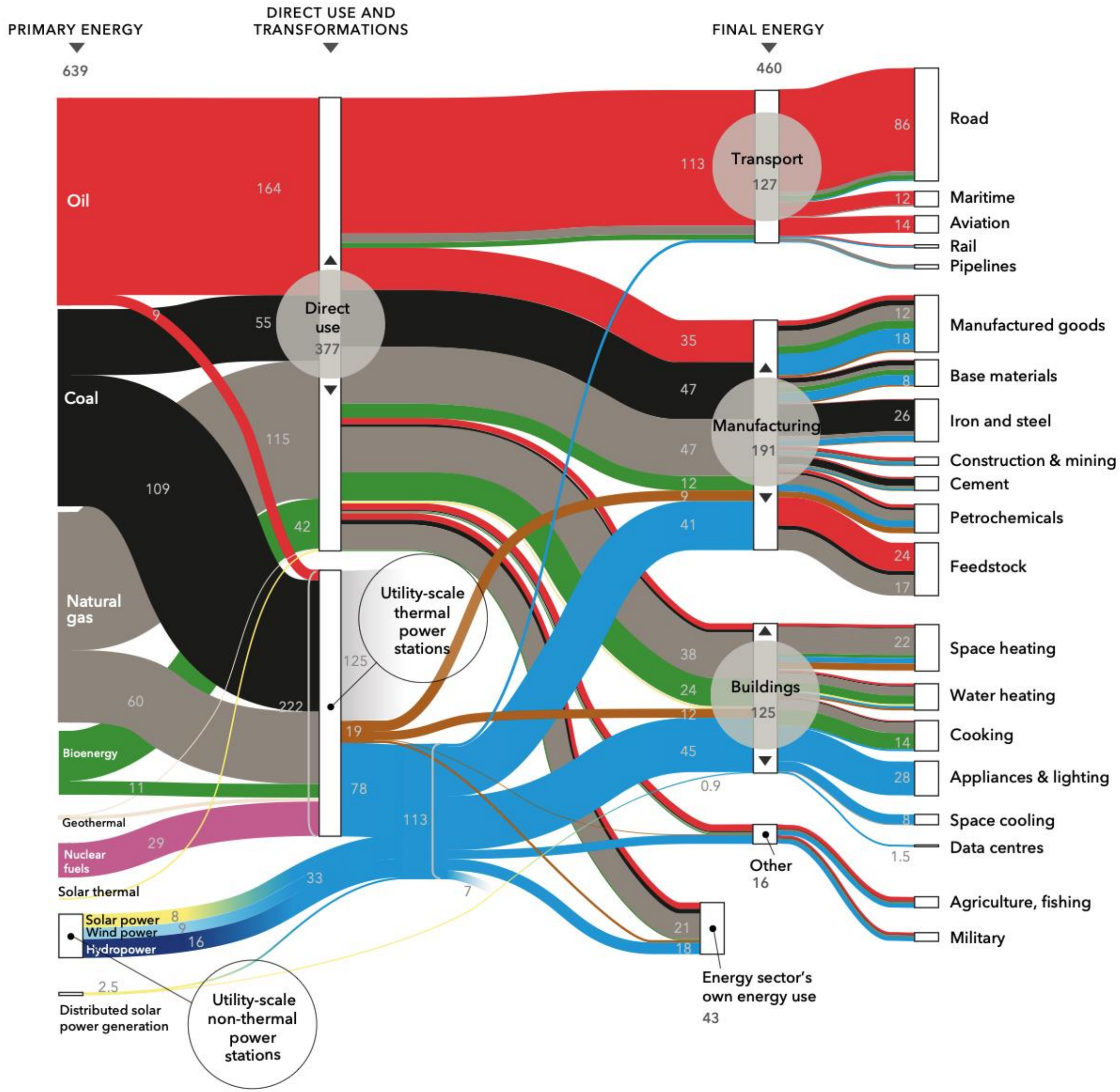
Global CO_2 emissions from transport

This is based on global transport emissions in 2018, which totalled 8 billion tonnes CO_2.
Transport accounts for 24% of CO_2 emissions from energy.



OurWorldinData.org – Research and data to make progress against the world's largest problems.
Data Source: Our World in Data based on International Energy Agency (IEA) and the International Council on Clean Transportation (ICCT).

https://ourworldindata.org/co2-emissions-from-transport



DNV: Energy transition outlook 2025. https://www.dnv.com/energy-transition-outlook/

Strategies for sustainable transportation

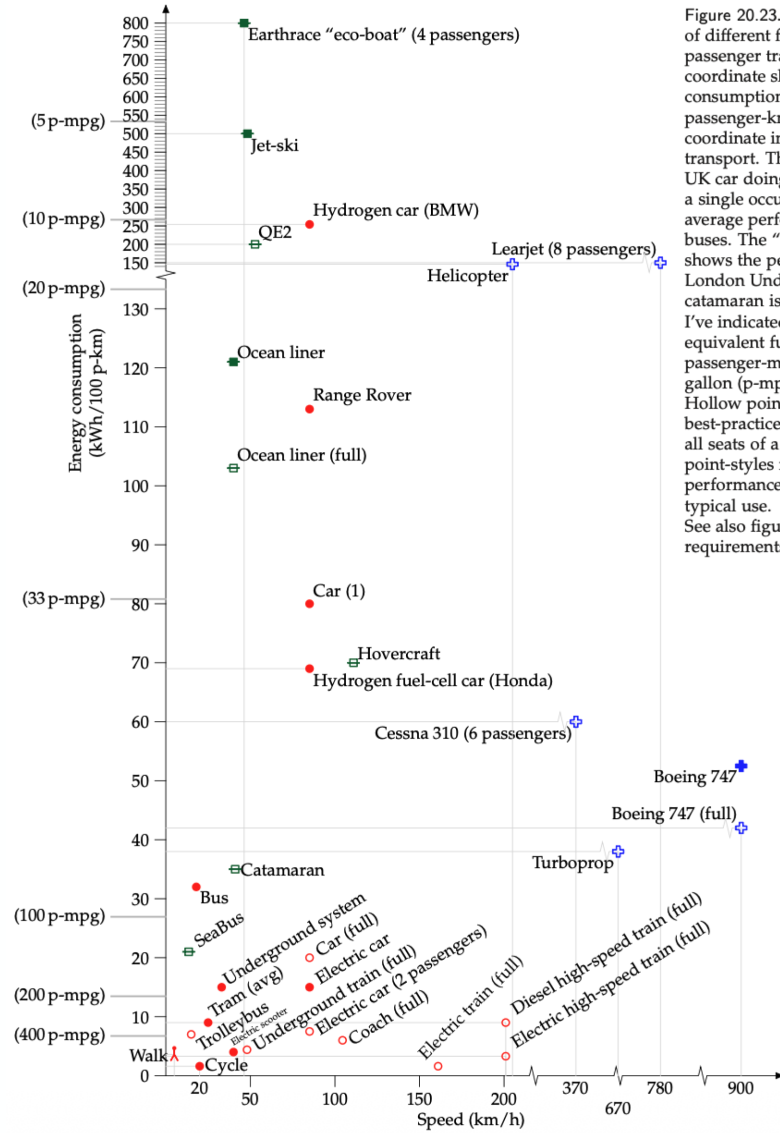


Figure 20.23. Energy requirements of different forms of passenger transport. The vertical coordinate shows the energy consumption in kWh per 100 passenger-km. The horizontal coordinate indicates the speed of the transport. The "Car (1)" is an average UK car doing 33 miles per gallon with a single occupant. The "Bus" is the average performance of all London buses. The "Underground system" shows the performance of the whole London Underground system. The catamaran is a diesel-powered vessel. I've indicated on the left-hand side equivalent fuel efficiencies in passenger-miles per imperial gallon (p-mpg).
Hollow point-styles show best-practice performance, assuming all seats of a vehicle are in use. Filled point-styles indicate actual performance of a vehicle in typical use.
See also figure 15.8 (energy requirements of freight transport).

What is efficiency?

Energy required to move a vehicle

Adding kinetic energy + overcoming resistance continuously

Accelerating a car from standstill 0 km/h to **50 km/h**

$$W_{total} = \Delta KE + W_{drag} + W_{roll} \qquad \Delta KE = \frac{1}{2}mv^2$$

Assume the car is Tesla Model 3 with two passenger (1750 kg + 70 kg + 70 kg = **1900 kg**)

$$\Delta KE = \frac{1}{2} \cdot 1900kg \cdot (13.89\, {}^{m}/_{s})^2 = 183256\, \frac{kg \cdot m^2}{s^2} = \mathbf{183\ kJ}$$

Now calculate the rolling resistance

$$W_{roll} = C_{roll} \cdot m \cdot g \cdot d \qquad d = \frac{1}{2}at^2$$

Assume coefficient of rolling resistance $\mathbf{C_{roll} = 0.01}$ and you accelerate from 0 to 50 km/h in **10 sec**

$$W_{roll} = 0.01 \cdot 1900kg \cdot 9.81\, {}^{m}/_{s^2} \cdot \left(\frac{1}{2} \cdot \left(\frac{13.89\, {}^{m}/_{s} - 0\, {}^{m}/_{s}}{10\, s}\right) \cdot (10s)^2\right) = 19kg \cdot 9.81\, {}^{m}/_{s^2} \cdot 69.5m = \mathbf{13\ kJ}$$

Finally, work needed to overcome the aerodynamic resistance

$$W_{drag} = \frac{1}{2} \cdot \rho \cdot C_{drag} \cdot A \cdot a \cdot d^2$$

The car is advertised to have air drag coefficient $\mathbf{C_{drag} = 0.22}$ and its cross-sectional area is roughly equal to $\mathbf{2.2\ m^2}$

$$W_{drag} = \frac{1}{2} \cdot 1.225\, {}^{kg}/_{m^3} \cdot 0.22 \cdot 2.2m^2 \cdot 1.389\, {}^{m}/_{s^2} \cdot (69.5m)^2 = \mathbf{2\ kJ}$$

$$W_{total} = 183 + 13 + 2 = \mathbf{200\ kJ} = \mathbf{55\ Wh} = \mathbf{790\ Wh/km}$$

Energy required to maintain vehicle's speed for 1 km

Overcome resistance continuously

Travelling 1 km at a constant speed of 50 km/h

$$W_{total} = W_{drag} + W_{roll}$$

$$W_{roll} = 0.01 \cdot 1900kg \cdot 9.81\,{}^{m}/_{s^2} \cdot 1000m = \mathbf{186\ kJ}$$

$$W_{drag} = \frac{1}{2} \cdot \rho \cdot C_{drag} \cdot A \cdot V^2 \cdot d$$

$$W_{drag} = \frac{1}{2} \cdot 1.225\,{}^{kg}/_{m^3} \cdot 0.22 \cdot 2.2m^2 \cdot (13.89\,{}^{m}/_{s})^2 \cdot 1000m = \mathbf{57\ kJ}$$

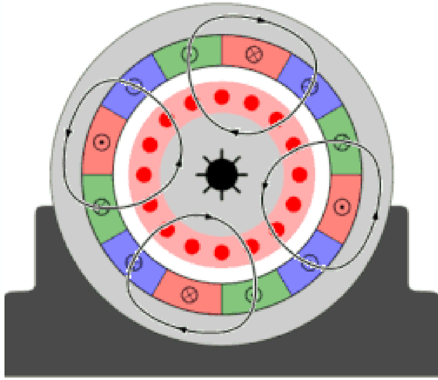
$$W_{total} = 186 + 57 = \mathbf{243\ kJ} = \mathbf{67.5\ Wh}$$

Total energy input over 69.5 m and 1 km

$$\mathbf{67.5\ Wh} + \mathbf{55\ Wh} = \mathbf{122.5\ Wh}$$

Regenerative breaking

Motors in electric vehicles can also generate electricity if you supply kinetic energy



$$E_{regen} = KE - W_{roll} - W_{drag}$$

$$E_{regen} = 183 - 13 - 2 = 168\ kJ$$

Not all that energy can be recuperated due to losses – motor (90%), inverter (90%), battery charging (90%) = 73%

$$E_{regen} = 168\ * 73\% = 123\ kJ = \mathbf{34\ Wh}$$

Total energy for EV

$$\mathbf{67.5\ Wh + 55\ Wh = 122.5\ Wh - 34\ Wh = 88.5\ Wh}$$

Energy efficiency

Internal combustion engine (ICE) vs electric vehicles (EV)

EV = 150 Wh/km

Modern ICE = 7 l/100km = 620 Wh/km

EV efficiency = 88.5 / 150 = 60%

40% is lost due to drive train, climate control, infotainment, etc.

ICE efficiency = 122.5 / 620 = 20%

80% is lost due to Carnot efficiency, drive train, climate control, infotainment, etc.

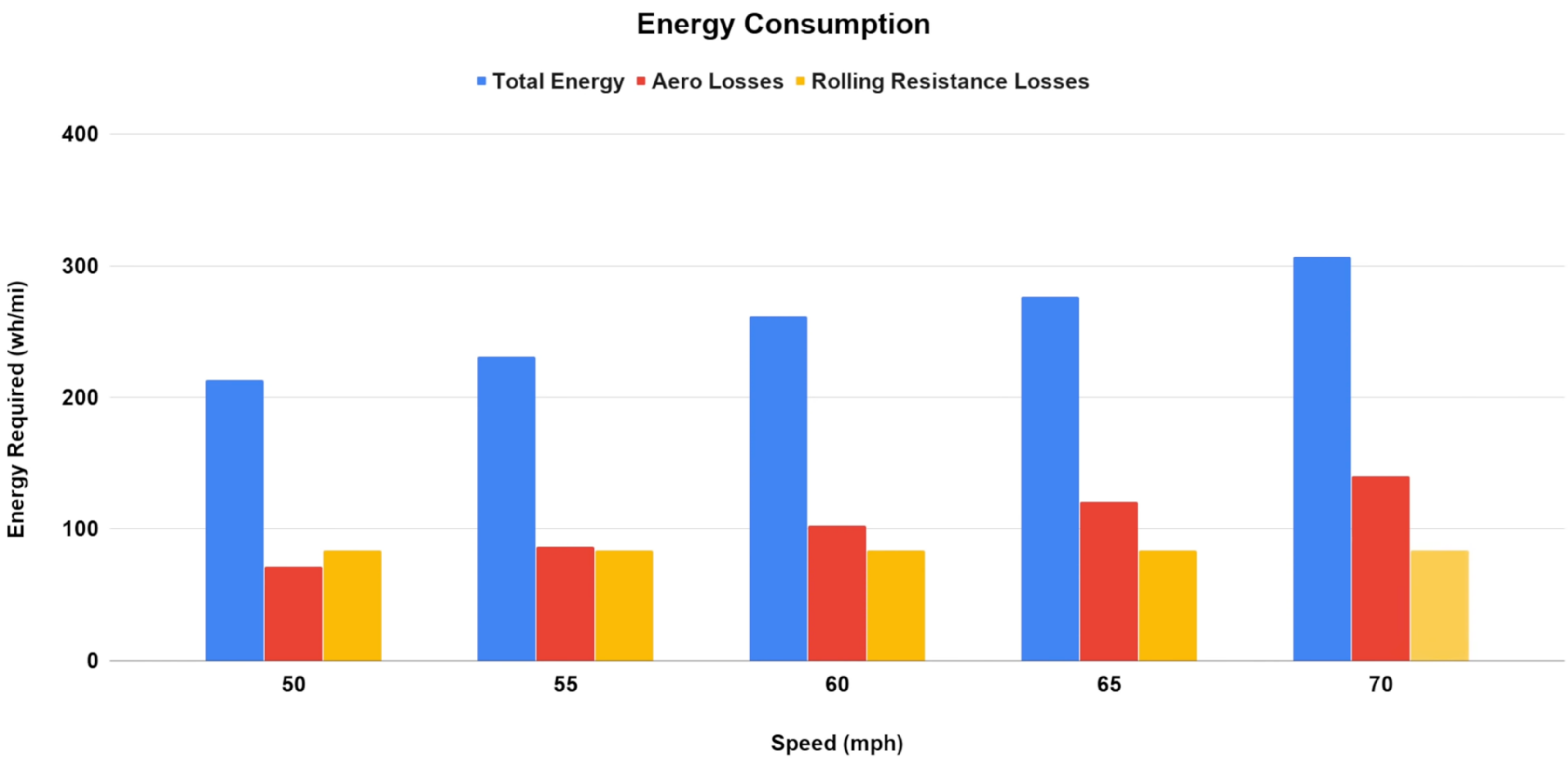
60/20 = 3 x more efficient

620/150 = 4.2 x more efficient

Real world tests

Traveling on a highway at constant speed

Average efficiency ~70%



Are Teslas Really That Efficient? – YouTube

Travelling 100 km – ICE vs EV

ICE = 7 litres / 100 km * 100 km = 7 litres = 62 kWh (50 $kgCO_2$)

EV = 150 Wh / km * 100 = 15 kWh

Diesel η 45%

39 kWh
31 $kgCO_2$

Transmission η 95%

17.5 kWh

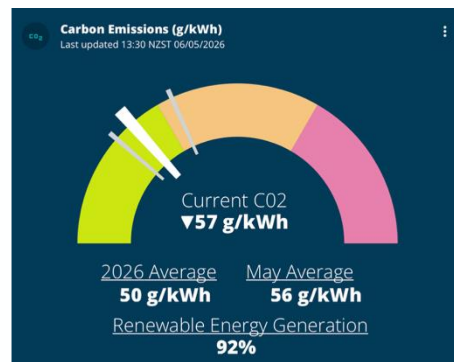
Charging η 90%

16.6 kWh

15 kWh

Coal η 40%

44 kWh
40 $kgCO_2$



17.5 * 50 = 0.88 $kgCO_2$

Travelling 100 km – ICE vs EV – worst case

ICE = 6 litres / 100 km * 100 km = 6 litres = 53 kWh (42.4 kgCO_2)

EV = 200 Wh / km * 100 = 20 kWh (33% worse)

Diesel η 45%

52 kWh
41.6 kgCO_2

Transmission η 95%

23.4 kWh

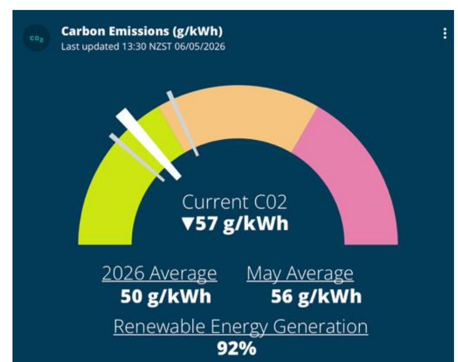
Charging η 90%

22.2 kWh

20 kWh

Coal η 40%

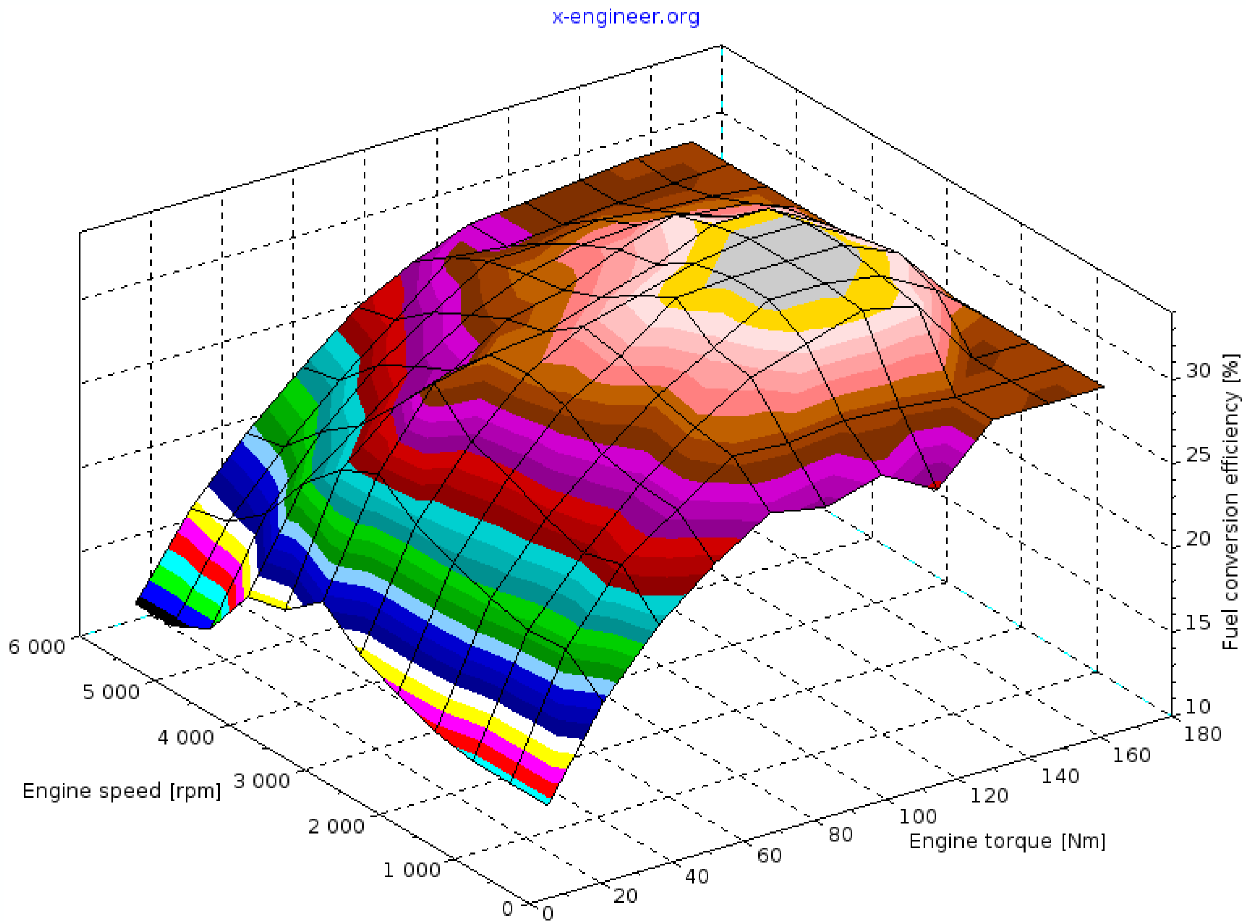
59 kWh
53 kgCO_2



23.4 * 50 = 1.2 kgCO_2

Stationary ICE vs vehicle

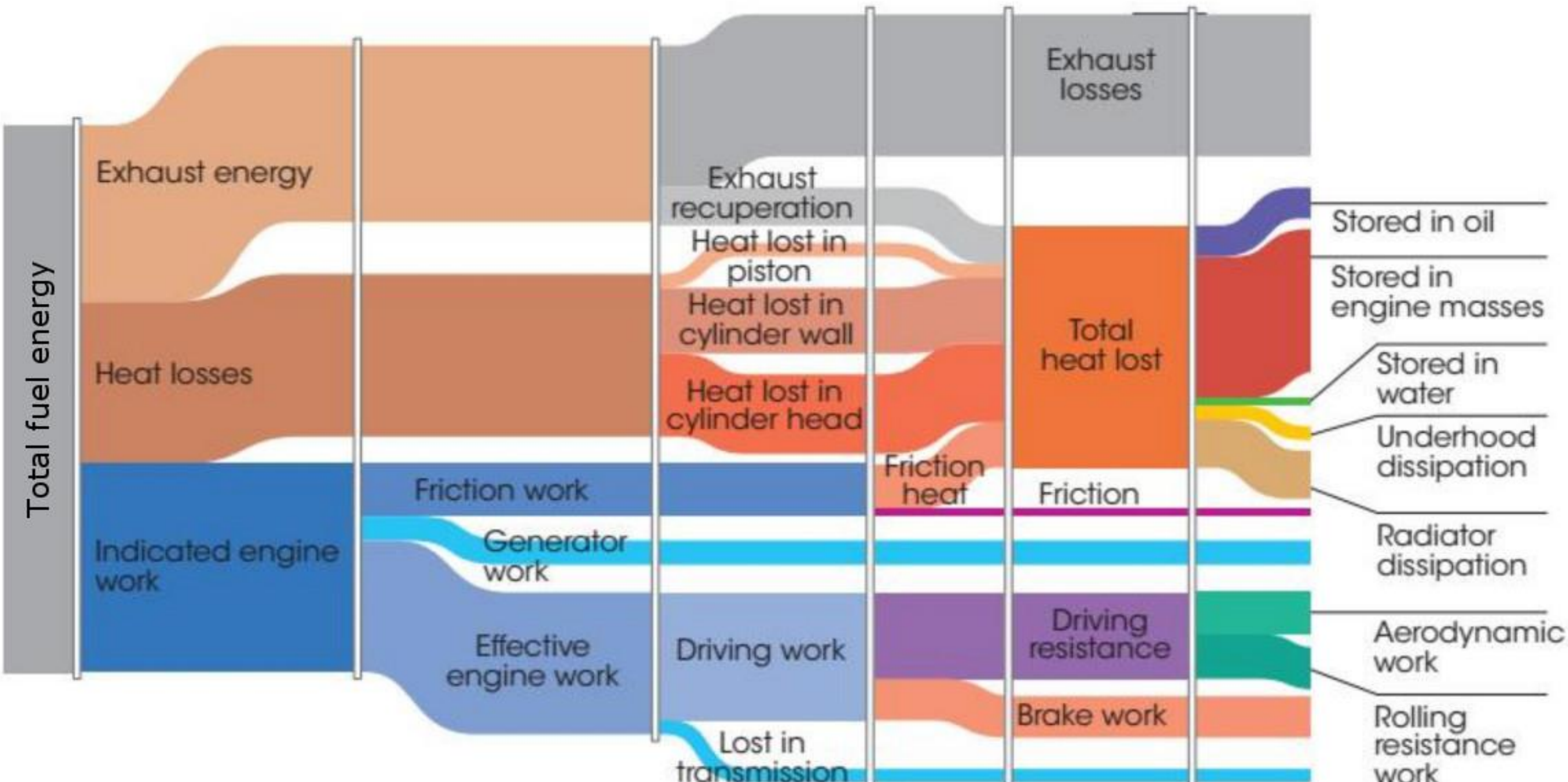
Stationary internal combustion generators maintain constant RPM and maximise energy output per fuel



https://x-engineer.org/brake-specific-fuel-consumption-bsfc/

Losses in an ICE vehicles

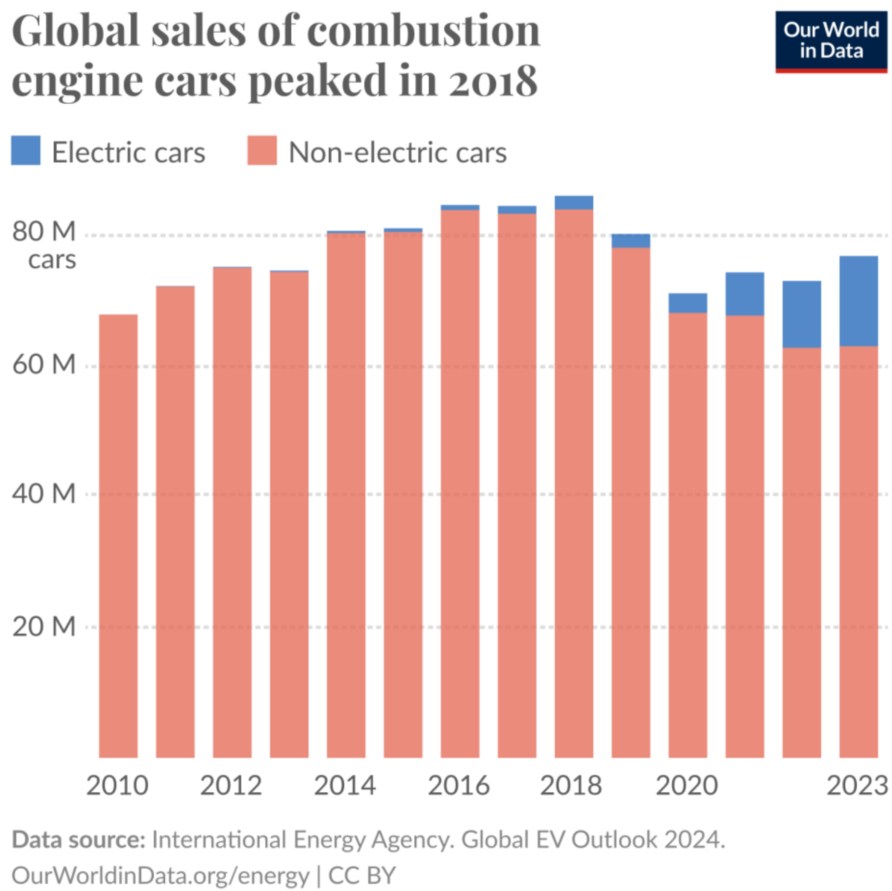
Very little goes into W_{drag} and W_{roll}



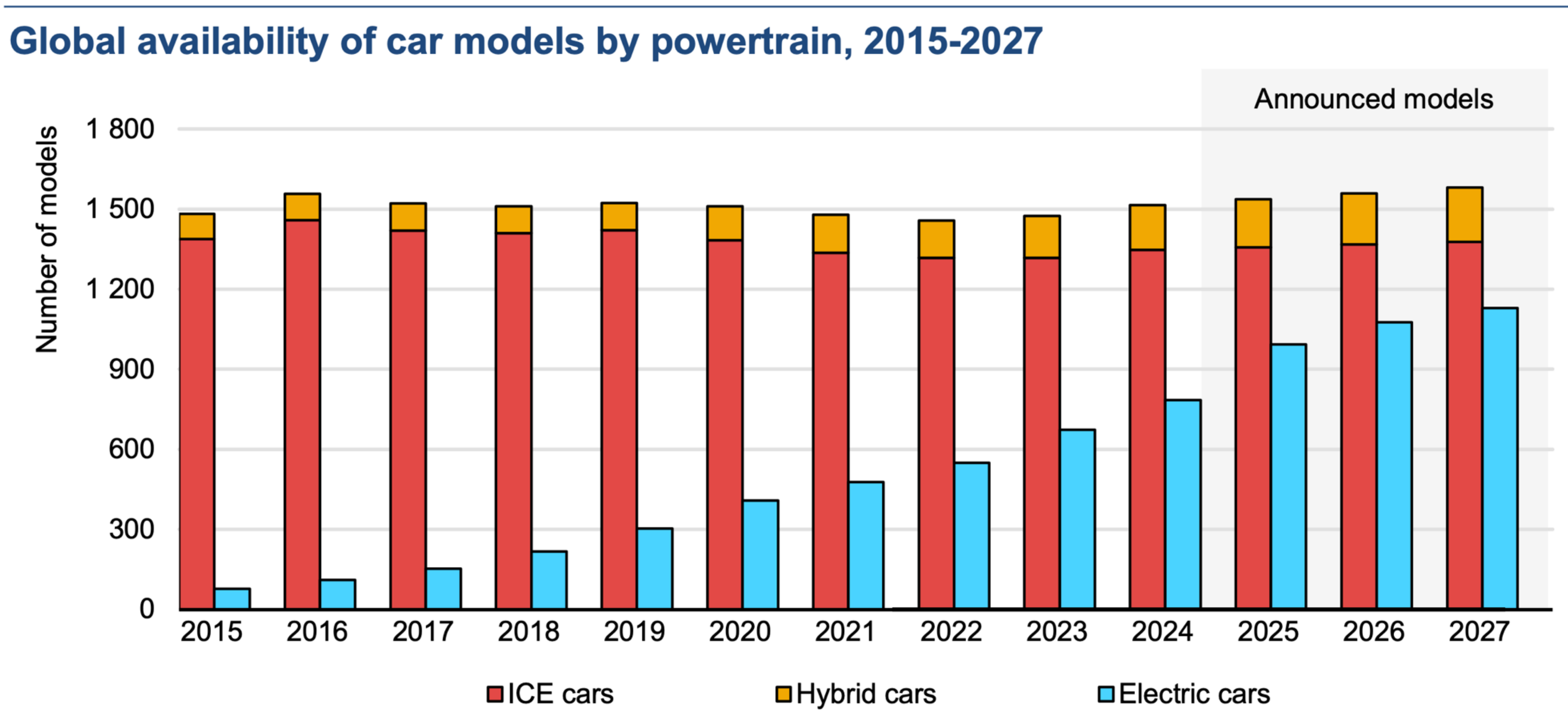
https://x-engineer.org/brake-specific-fuel-consumption-bsfc/

ICE sales peaked in 2018

Implications?

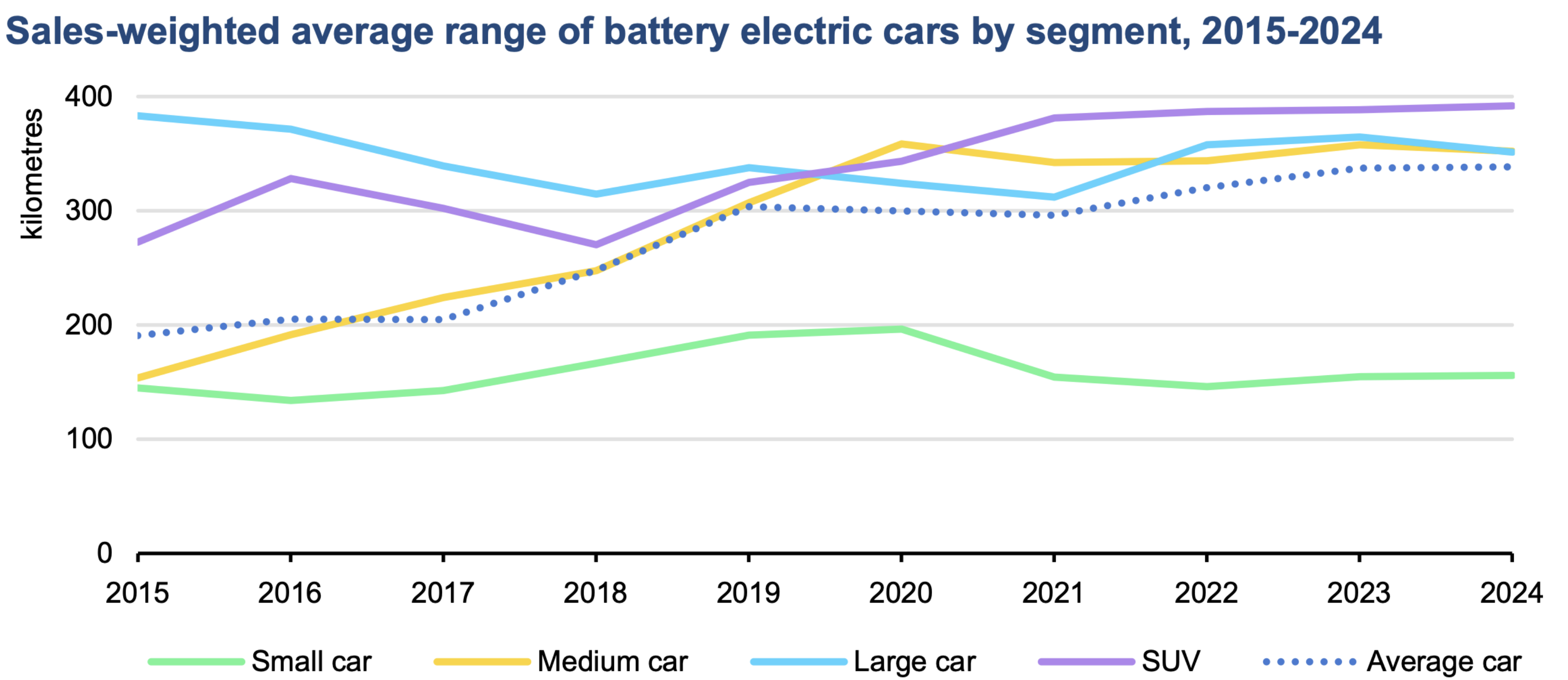


“I don’t like the options on the market”



IEA. Global EV Report. 2025

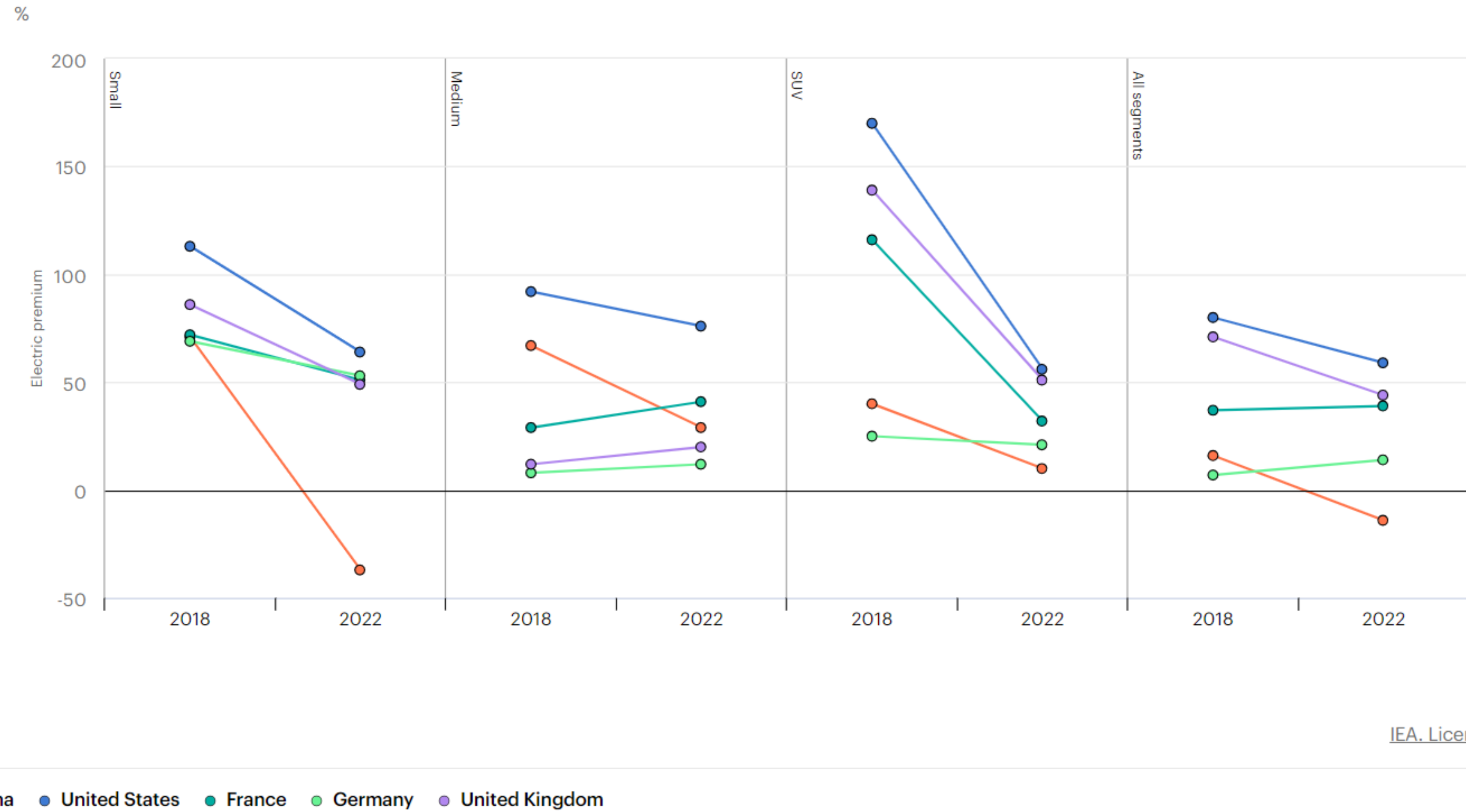
Range anxiety



IEA. Global EV Report. 2025

EVs to reach sticker-price parity - with subsidies

Subsidies often designed to tackle EV premium

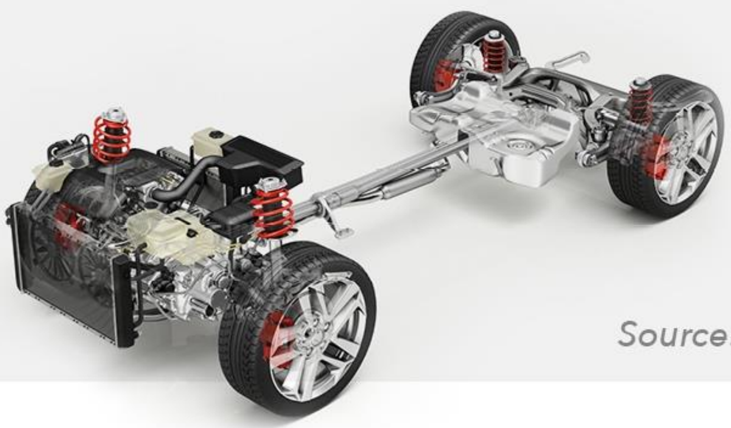


IEA: Global EV Outlook. 2024. https://www.iea.org/reports/global-ev-outlook-2024/trends-in-electric-cars

ICE Powertrain

1,400 Components

Engine, Exhaust, Transmission/Drivetrain



BEV Powertrain

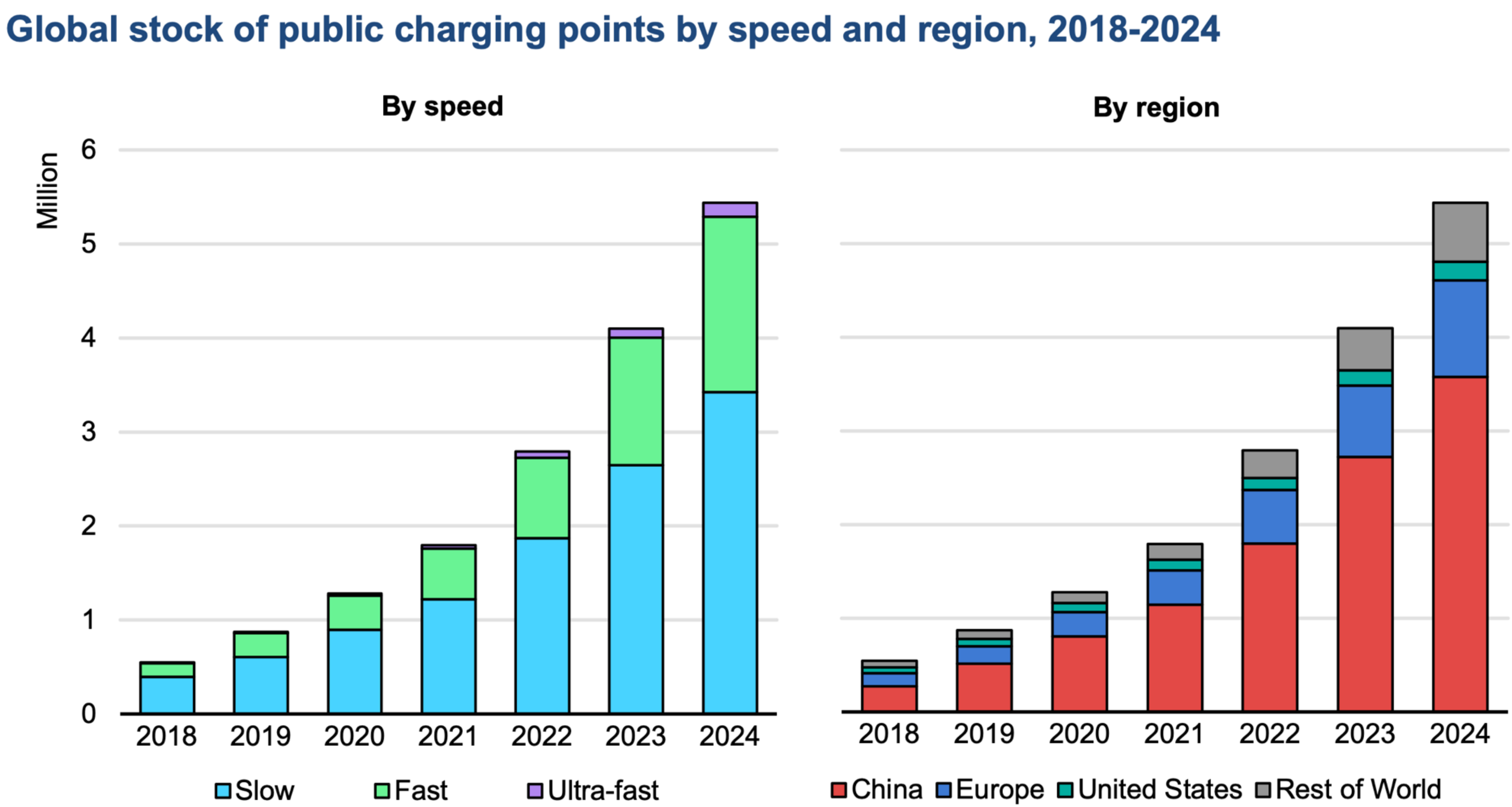
200 Components

Electric Motor (+ Power Electronics), Battery Pack

Source: Roland Berger

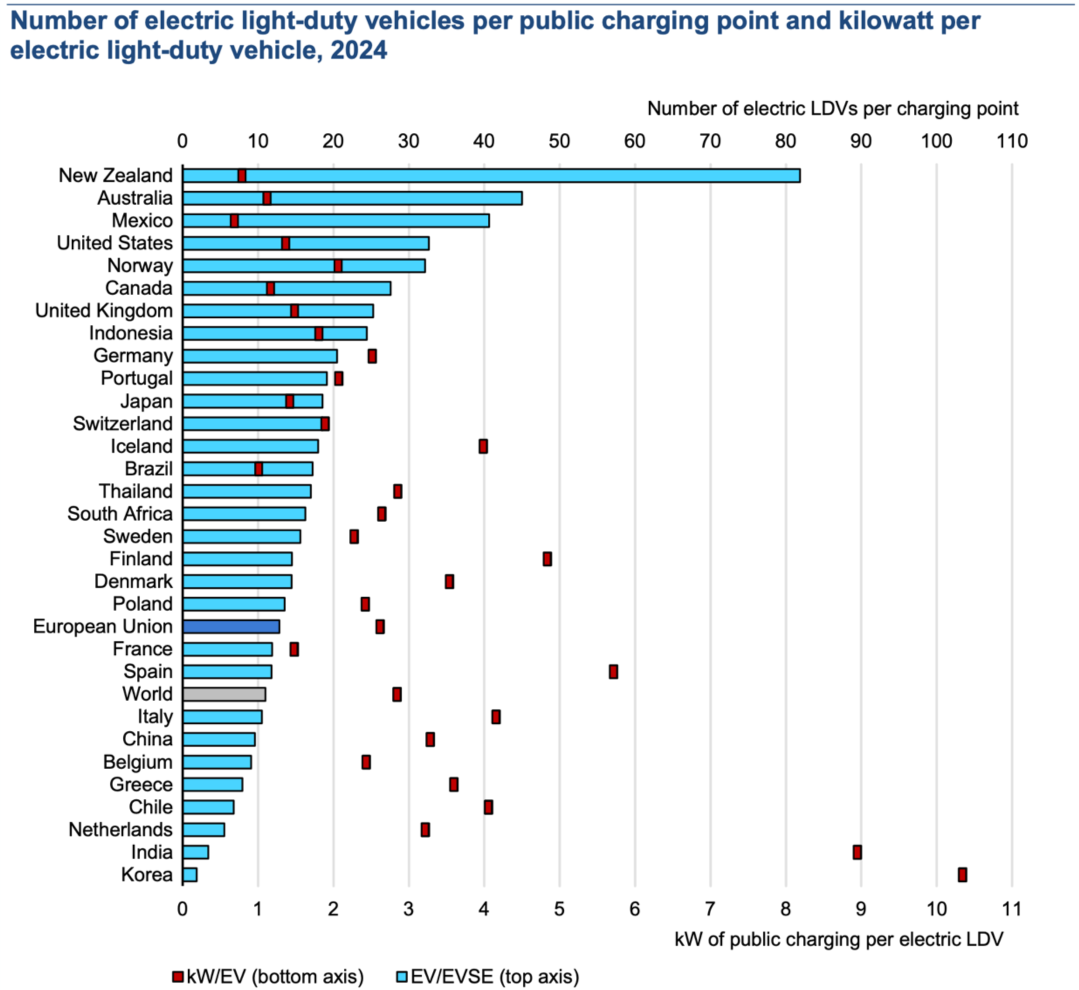
Public chargers

Growth is accelerating



IEA. Global EV Report. 2025

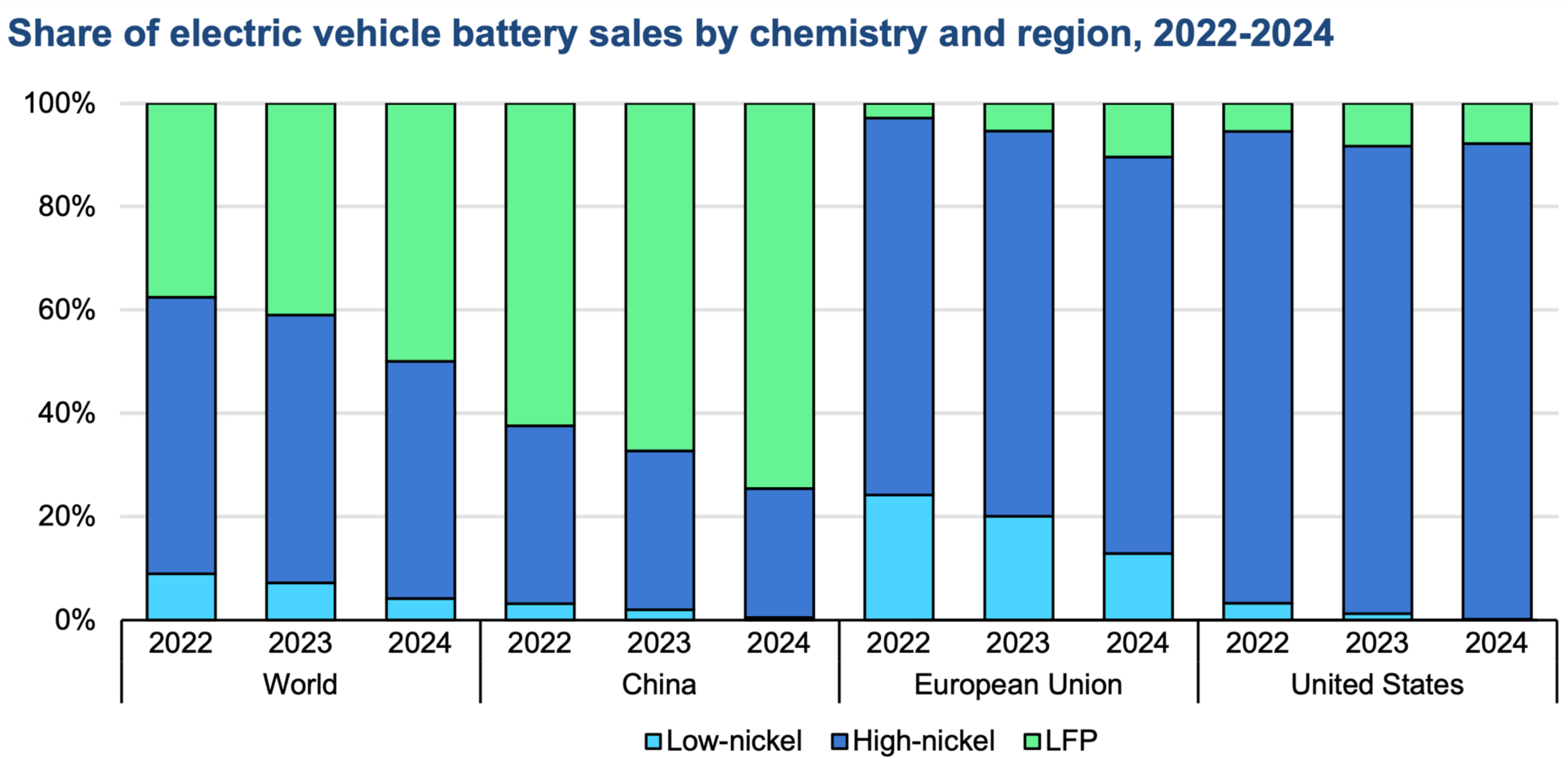
EVs per public charger



IEA. Global EV Report. 2025

Battery chemistry

Mostly lithium iron phosphate



IEA. Global EV Report. 2025

Flow of cars

Dominated by China – downstream from battery domination